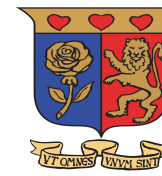


Wireless Networks and WLANs

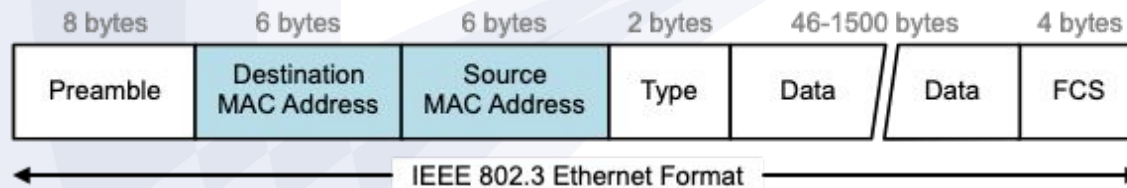
IEEE 802.11 Frames, Addressing and Security

Ethernet Frame Format

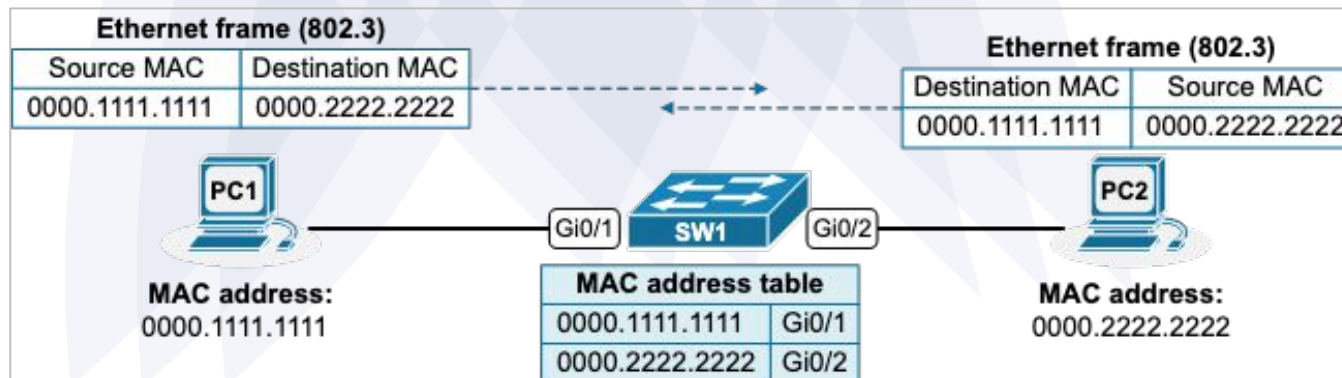


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- The diagram shows a high-level overview of the structure of an Ethernet frame used in switched networks. Notice that it has two layer 2 addresses: Source MAC and Destination MAC. This is important because wireless frames have more than two MAC addresses!

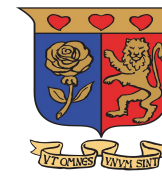


- In traditional Ethernet networks (IEEE 802.3), devices send and receive data in a structured frame format, as shown in the diagram below. Each frame contains two MAC addresses: a source MAC address and a destination MAC address. The source address is not used for frame delivery but tells the recipient where to send a reply.

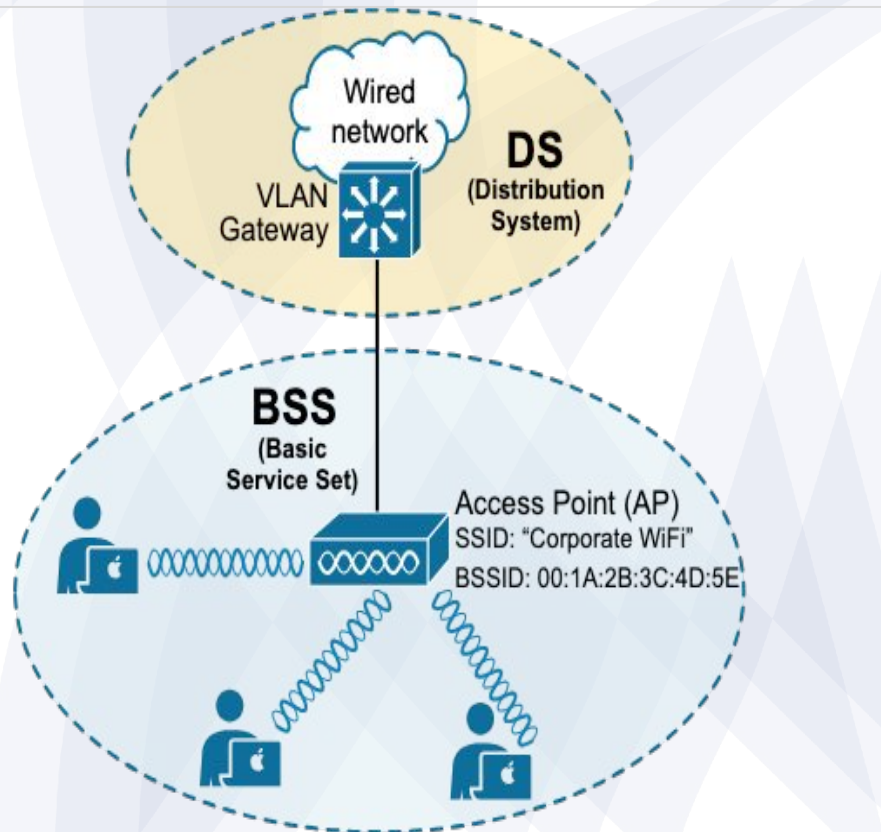


- When a device receives a frame with its own MAC address as the destination, it accepts and processes the frame.
- Imagine a sender and receiver are connected through a network switch. The switch forwards frames between them based on MAC addresses but doesn't modify the addresses in the frames.
- Also, notice that the end devices aren't aware of the switch—they simply forward frames based on its routing/ARP table.
- They don't have visibility into the switched network and don't know whether they are connected to a switch, hub, router, or another network device.

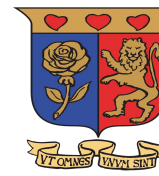
IEEE802.11 Frame: Recall BSS



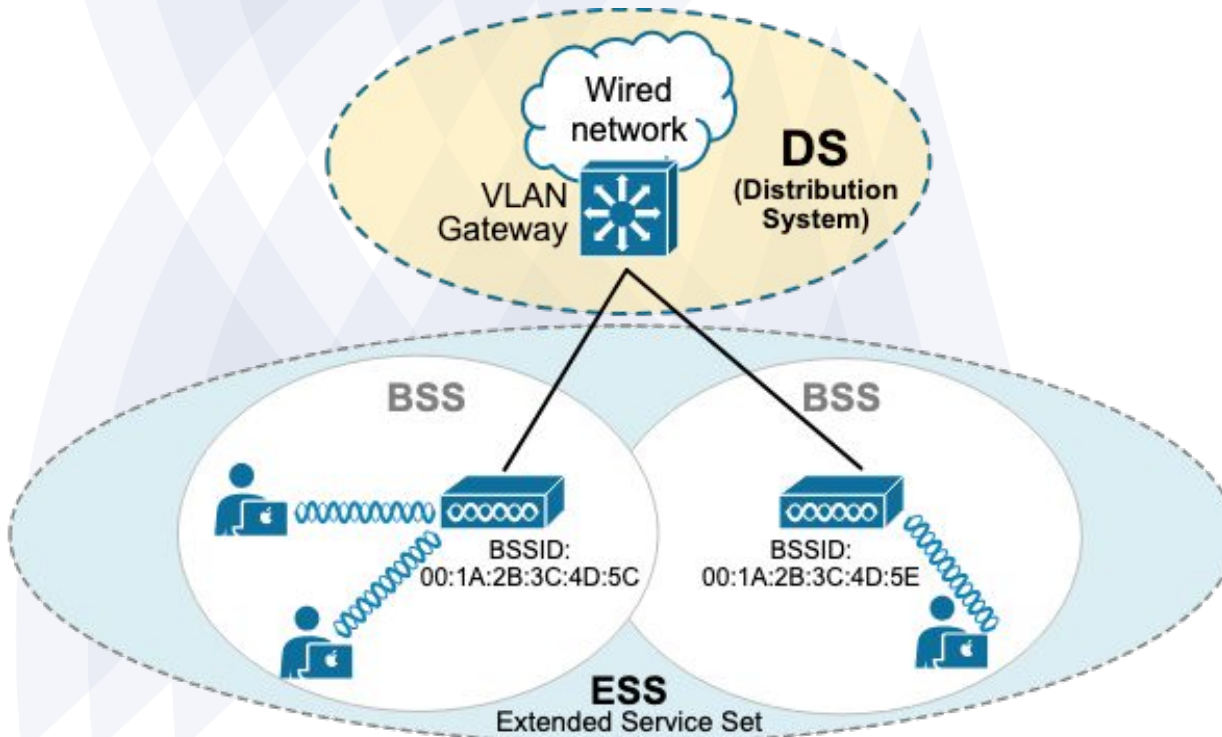
- In an 802.11 wireless network, the access point (AP) plays an active role in managing the communication between clients. Wireless devices must first associate with an AP to join the network.
- The AP acts as a central hub. All data passes through it. Clients are not allowed to communicate directly without using the AP. The word "hub" is intentionally used because it indicates that all communication is half-duplex (similar to ethernet communication via a hub).
- In wireless networks, the Basic Service Set (BSS) is the smallest unit of a Wi-Fi network. It consists of an access point and the wireless clients connected to it. Each BSS has a unique identifier called the BSSID, which is the MAC address of the AP.



IEEE802.11 Frame: Recall ESS and DS

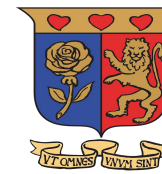


- Multiple BSSs connected to create a larger area with wifi coverage, form an Extended Service Set (ESS).
- ESS is a larger Wi-Fi network made up of multiple BSSs connected through a wired network. This allows clients to move between APs while staying on the same network, enabling roaming. The entire ESS is identified by a single SSID (the human-friendly network name).

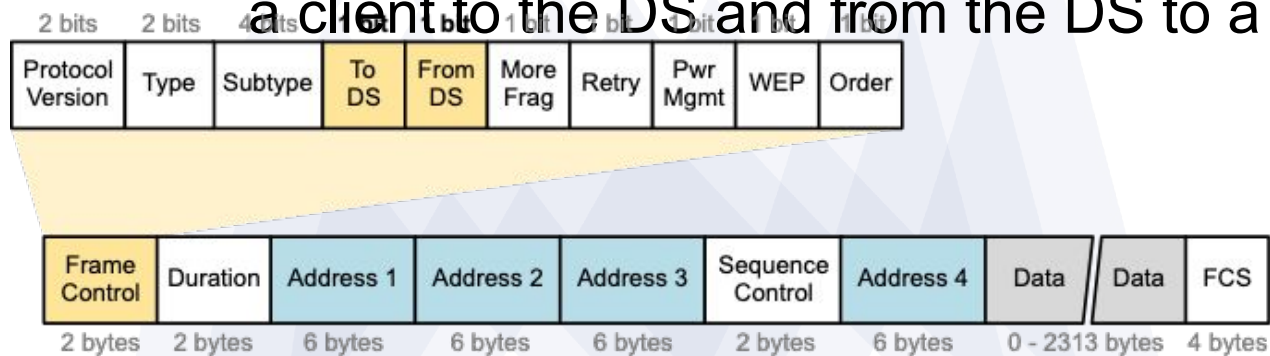


- Multiple BSSs connected to create a larger area with wifi coverage, form an The DS is the wired network that connects multiple APs in an ESS.
- It allows APs to communicate with each other and forward client traffic to other networks, such as the Internet or internal servers.
- The DS is usually the LAN side of the AP.

IEEE802.11 Frame: Frame Structure and Direction

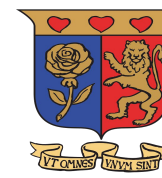


- The diagram shows the structure of an 802.11 frame. Notice that it starts with a Frame Control field (in yellow). It defines the frame type and its direction within the network.
- In most cases, wireless frames travel between clients and the AP, which then forwards them to the distribution system (DS) and, most likely, to the Internet and back.
- The 802.11 frame header includes two direction bits, "To DS" and "From DS" (in yellow), to indicate whether the frame is being sent to or coming from the DS.
- At first glance, it may seem that frames can only move in one of two directions: from a client to the DS and from the DS to a client. However, there are cases where frames do not follow simple pattern. let's look at a few examples

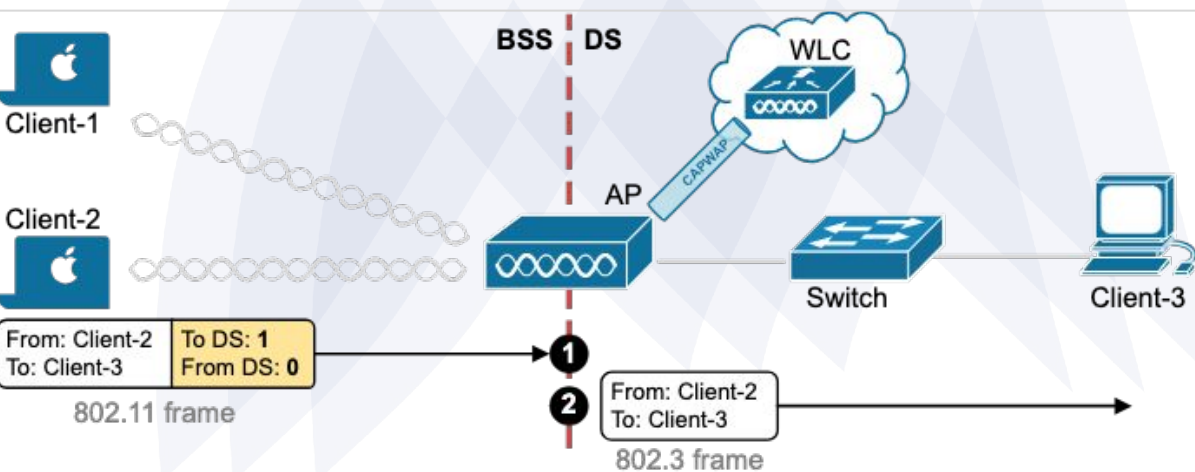
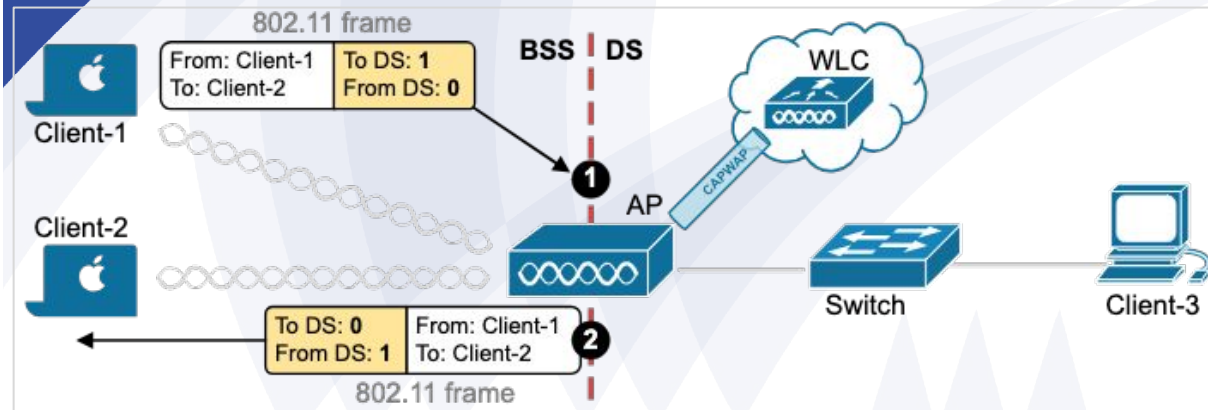


- In the example, two clients communicate with each other via the access point (even though both clients are connected to the same AP, in the split-MAC architecture (which is most common in large organizations) means traffic goes to the wireless controller and back via the CAPWAP tunnels).

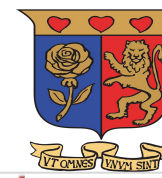
IEEE802.11 Frame: Frame Structure and Direction



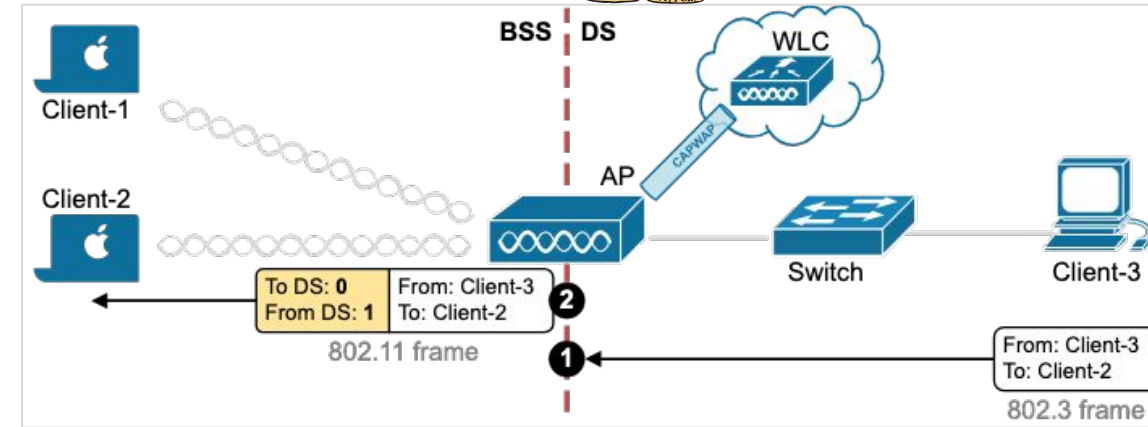
- **Step 1:** When Client-1 sends wireless frames, it sets the "To DS" bit to 1 and the "From DS" bit to 0. The frame is thus destined for the Distribution System for switching/routing to the correct destination. (Client-1 doesn't know where Client-2 is.)
- **Step 2.** If this is a split-MAC architecture, the frame from Client-1 reaches the AP and then the wireless controller via the CAPWAP tunnels. The controller switches the frame back to the access point. Before the AP sends the frame to Client-2, it changes the directions - it sets the "To DS" bit to 0 and the "From DS" bit to 1, as shown).



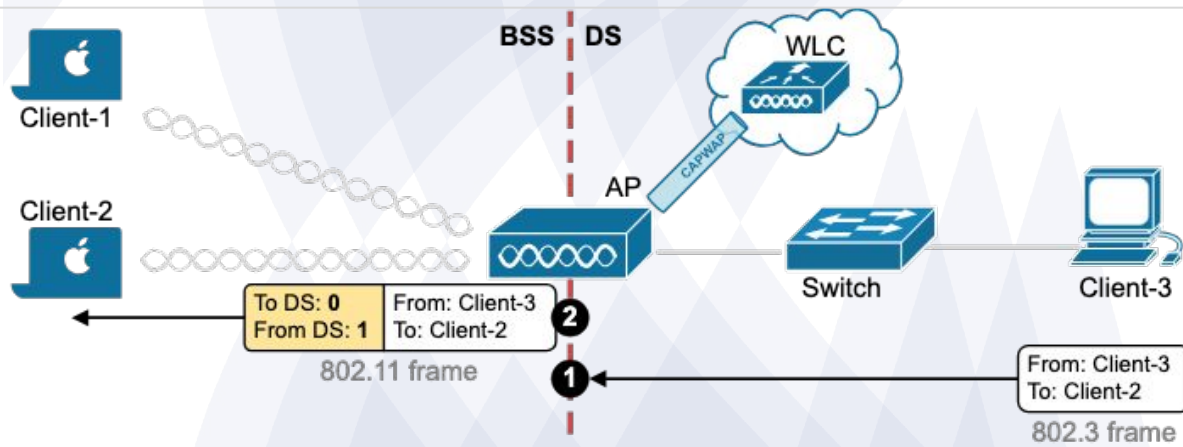
IEEE802.11 Frame: Frame Structure and Direction



- Consider when Client-2 sends data to Client-3, which is connected somewhere on the wired side of the network. Client-2 sets the directions as shown.



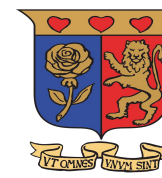
- When Client-3 responds, the AP changes direction in the 802.11 frame, as shown in step 2. It sets the "To DS" bit to 0 and the "From DS" bit to 1 to indicate that the traffic is coming from the Distribution System.



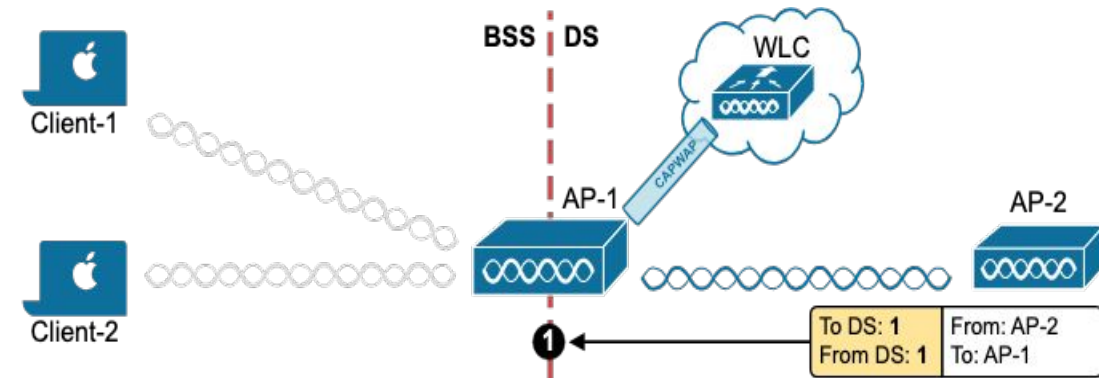
If there are only two possible directions in the network (to DS and from DS), a single bit should be adequate. For example, set to 0 for one direction and set to 1 for the reverse direction. Why does the frame header use two bits?

The answer is that there are special cases where data is not just going between devices and the DS but somewhere else, leading to four possible direction types.

IEEE802.11 Frame: Frame Structure and Direction



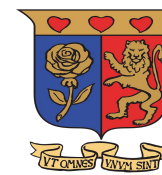
- One special case occurs when an AP sends a broadcast frame to all connected clients, meaning the frame does not originate from the DS.
- Another example is when a client sends a management frame directly to the AP, making the AP itself the destination.
- A different scenario arises in mesh networks, where APs communicate wirelessly with each other over backhaul links.
- Since these links are not part of a basic service set (BSS) or the DS, both direction bits in the frame header are set to 1, indicating that the frame is traveling between APs rather than between clients or the wired network, as shown in the diagram



If there are only two possible directions in the network (to DS and from DS), a single bit should be adequate. For example, set to 0 for one direction and set to 1 for the reverse direction. Why does the frame header use two bits?

The answer is that there are special cases where data is not just going between devices and the DS but somewhere else, leading to four possible direction types.

IEEE802.11 Frame: Frame Structure and Direction

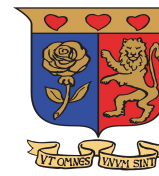


- The table summarizes the different combinations of frame direction. Most of the time, the wireless data moves from the BSS to the DS system or vice versa.

Frame Control : To DS	Frame Control: From DS	Use case
0	0	An AP itself (not the DS) is the source or destination of the frame. For example, an AP sends a control frame to all associated devices.
0	1	An 802.11 frame is sent from the DS to a client.
1	0	An 802.11 frame is sent from a client to the DS.
1	1	An AP relay a frame to another AP over a wireless backhaul link. The link is neither in the BSS nor in the DS.

- The From DS and To DS bits in the 802.11 header are important because they indicate the direction of the wireless data within the wireless network. These bits help devices understand how to handle and forward frames properly. The process is crucial for directing traffic correctly, ensuring frames reach their intended destination, and supporting advanced wireless capabilities like roaming, mesh networks, and split-MAC deployments.
- The key point to understand from this lesson is that Ethernet (802.3) frames have a simple structure with two MAC addresses: source and destination. In contrast, wireless (802.11) frames are more complex and can contain up to four MAC addresses.

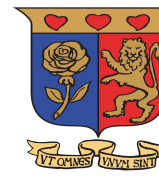
IEEE 802.11 Frame Addressing



- It is essential to understand why 802.3 Ethernet frames have only two address fields: Source MAC and Destination MAC while 802.11 Wireless frames have four address fields that are used differently depending on the originator and the direction of the frame.
- An 802.11 frame header contains four address that have generic names: Address1, Address2, Address3 and Address4. They are not listed as "Source MAC" and "Destination MAC," as in the standard 802.3 Ethernet header. The reason is that the address fields are used differently in different situations. Their value depends on where the frame is going and who the originator is.
- Since 802.11 frames always travel from a radio transmitter to a radio receiver, the first two addresses are: Address_1 always holds the RA (Receiver Address) and Address_2 always holds the TA (Transmitter Address).
- These are not identical to the source and destination Ethernet MAC addresses. The transmitter address is the MAC of the radio device that sends the RF signal into the air. The receiver address is the MAC of the radio device that should receive the radio signal (not the ultimate layer 2 destination).

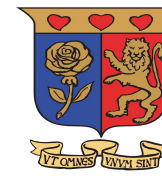


IEEE 802.11 Frame Addressing



- Typically, most of the wireless traffic goes from a client to the Internet. In that case, the Transmitter is the client, so Address_2 is the client's MAC address. The receiver of the radio signal is the access point (AP), so Address1 is the AP's MAC address, which is the BSSID.
- The frame is destined to the subnet's default gateway because the traffic goes to the Internet, making Addresses 1 and 2 are insufficient. That's why the frame needs at least one more field for the ultimate layer 2 destination.
- Address_3 holds the Destination Address (DA) if the receiver (usually the access point) is not the final layer 2 destination. For example, when a wireless client sends a frame to a device on the wired network (DS), the AP receives the frame and forwards it. Address_3 can also hold the Source Address (SA) if the transmitter (usually the access point) is not the original sender. For instance, when a wired device sends a frame to a wireless client, the AP transmits it over the air, but the frame originated from the wired device.
- Address_4 is used only when one AP is forwarding a frame to another AP over a wireless link
- Sometimes, frames need to pass through an intermediate wireless device before reaching their final destination. In such cases, address_3 and Address4 carries the original SA and DA, while Address1 and Address2 remain the RA and TA for the APs relaying the frame.

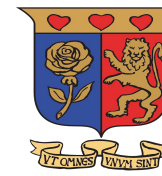
IEEE 802.11 Frame Addressing



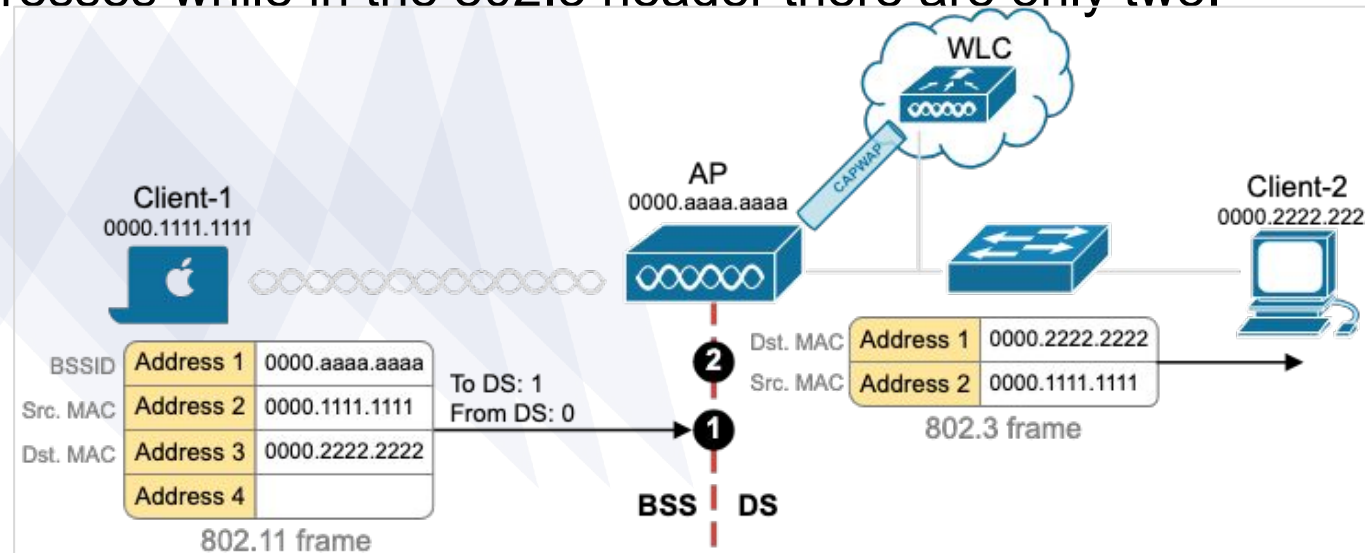
- It is enough to simply remember that there is a significant difference between 802.11 and 802.3 frames.
- Review the Table: First, keep in mind that 99.9% of the traffic will be either row two (DS to client) or row three (client to DS). In that case, addresses 1 and 2 are the MAC addresses of the client and the AP because they are the wireless transmitter and receiver.
- Traffic at the AP is forwarded toward the wired network. That's why Address 3 is required to store the ultimate layer 2 source/destination - typically the MAC address of the VLAN gateway.
- Rows 1 and 4 in the table are used relatively rarely. Row 1 is for control/management frames. Row 4 is only used if a frame must be relayed from one AP to another AP over a wireless backhaul link or mesh, which is not very common in enterprise wireless networks.

	To DS	From DS	Address1 (Receiver)	Address2 (Transmitter)	Address3	Address4
Management, Control or DTL frames	0	0	Dest MAC	Src MAC	BSSID	
DS to a client	0	1	Dest MAC	BSSID	Src MAC	
A client to DS	1	0	BSSID	Src MAC	Dest MAC	
Wireless mesh or bridge	1	1	Dest Radio	Src Radio	Dest MAC	Src MAC

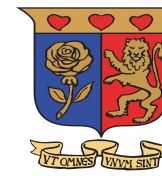
IEEE 802.11 Frame Addressing



- Using the table, the example matches the use case described in row three - a client to DS.
 - Address_1 is the wireless receiver address. In this case, it is the AP's MAC, which is, in fact, the BSSID.
 - Address_2 is the wireless transmitter address. In this case, it is the client's MAC address.
 - Address_3 is the ultimate layer 2 destination, which is Client-2's MAC address.
- Notice that once the frame goes out of the wireless network, the device that bridges it to the LAN converts it into an 802.3 Ethernet frame. Usually, this is the controller (WLC), but it may also be the access point if the organization uses FlexConnect or Autonomous APs.
- Notice how the address field differs between the different header types. In 802.11 there are three addresses while in the 802.3 header there are only two.



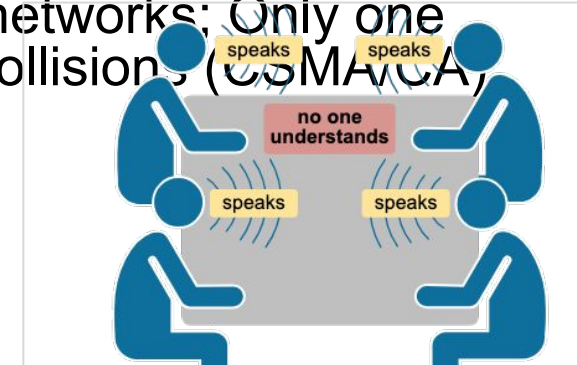
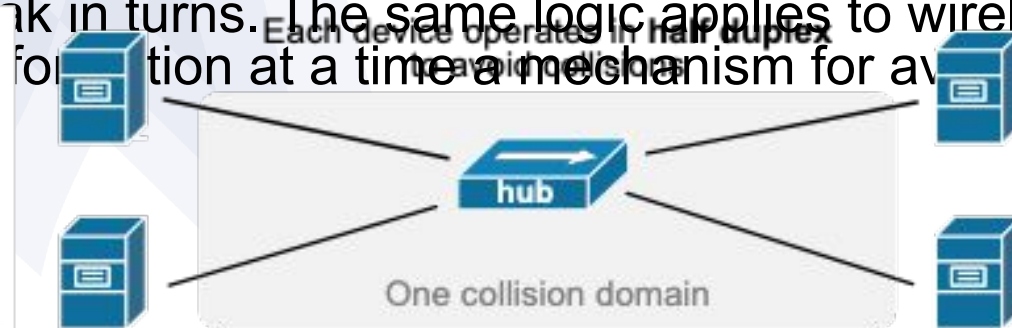
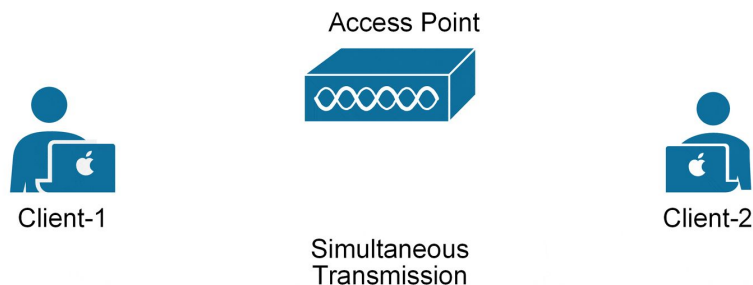
Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)



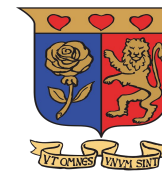
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- In shared networks like wireless or wired Ethernet, multiple devices transmitting simultaneously can cause collisions. Collisions corrupt data, waste time, and require retransmission, which wastes even more time.
- In wireless networks, the algorithm used tries to prevent/avoid collisions because of interference.
- All wireless devices inside a BSS share the same medium - a single RF channel based on the Wi-Fi standard they use. When multiple devices transmit at the same time, they interfere with each other, and this interference becomes more likely as more wireless devices are added. For example, two devices transmitting simultaneously on the same channel, the two RF signals cancel each other out, and the access point receives corrupted data.
- It is the same effect in an Ethernet LAN that is a single collision domain. Recall that devices connected to a hub are in the same collision domain. To use the media effectively, all hosts must operate in half-duplex mode, which means they can't send and receive data at the same time.

The solution is to make devices talk in turns. The same logic applies to wireless networks; Only one device can transmit at a time - a mechanism for avoiding collisions (CSMA/CA).



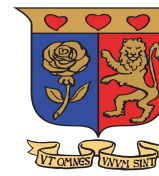
Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)



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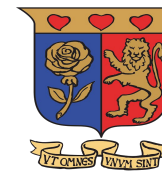
- Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) is a MAC layer protocol used in 802.11 wireless networks to ensure only one device transmits at a time.
 - **Carrier Sense (CS):** Expresses the idea that a device may only transmit data if the transmission medium is free. Therefore, every device constantly monitors the RF spectrum to "sense" if someone is currently transmitting.
 - **Multiple Access (MA):** The wireless medium is shared between all connected devices. Devices use the same frequency space. Therefore, they must follow a common set of rules to avoid interference.
 - **Collision Avoidance (CA):** Devices use a backoff mechanism to reduce the chances of two devices transmitting simultaneously.
- CSMA/CA tells each device to listen to the medium and "sense" if the channel is free/busy. If no other device is transmitting, it proceeds with the transmission. If the channel is busy, the device waits for a random amount of time—called the **back-off period**—before checking again. Once the backoff counter reaches zero and the channel is clear, the device sends its data. After receiving the transmission, the recipient sends back an acknowledgment (ACK) to confirm successful delivery.
- To further reduce collisions, an optional Request to Send/Clear to Send (RTS/CTS) mechanism can be used. With RTS/CTS, a device first requests permission from the access point before sending data. If approved, it proceeds with transmission, ensuring no other device transmits at the same time.

CSMA in Wired Vs Wireless Networks



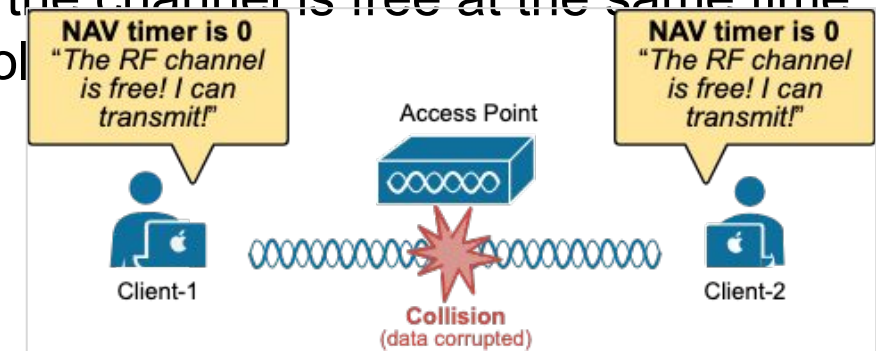
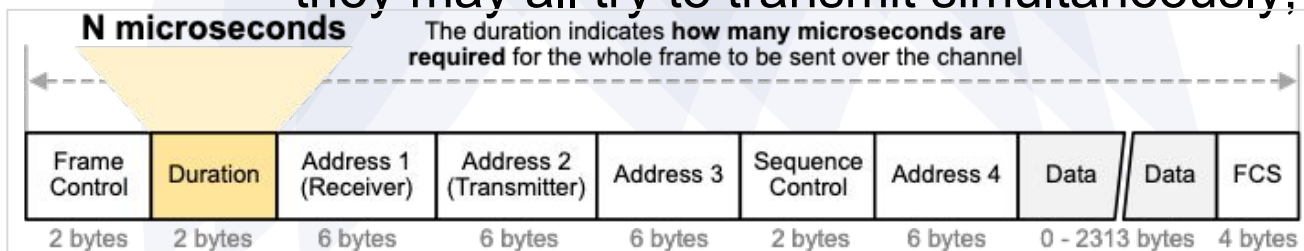
- CSMA works differently in wireless and wired networks because the transmission medium is different and has different constraints.
- In wired networks, devices can detect collisions because they can listen for electrical signals on the cable while transmitting. If a device detects an unexpected signal (caused by another device also transmitting), it recognizes a collision has occurred.
- In wireless networks, it is impossible to detect collisions because devices cannot listen and transmit simultaneously on the same frequency. When a device is transmitting, its own RF signal is much stronger than any incoming signal, making it impossible to "hear" other transmissions and detect a collision.
- Therefore, there are two different CSMA algorithms: CSMA/CA (Collision Avoidance) and CSMA/CD (Collision Detection). The main difference between CSMA/CA and CSMA/CD is how they handle collisions:
 - CSMA/CA (Wireless) tries to prevent collisions before they happen by making devices check if the channel is clear before transmitting.
 - CSMA/CD (Wired) allows collisions to happen but detects and resolves them by making devices stop and resend their data.

Carrier Sensing

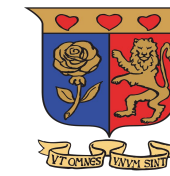


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- Wireless devices use two methods to check if the transmission medium (the wireless channel) is in use.
 - Physical Carrier Sense (Clear Channel Assessment - CCA) - When a wireless device is not transmitting, it listens to the channel for other transmissions. If it detects a signal, it assumes the channel is busy. High-bandwidth Wi-Fi devices, like those using 802.11n or 802.11ac, must check multiple channels at once. If a frame is overheard and contains the device's MAC address, it processes the frame. If not, it still recognizes that the channel is in use.
 - Virtual Carrier Sense (Network Allocation Vector - NAV) - The second method is virtual carrier sense, which relies on the Network Allocation Vector (NAV). Every transmitted frame includes a duration value that tells other devices how long the channel will be occupied. Devices that overhear the frame store this value in their NAV timer and wait until it reaches zero before attempting to transmit.
- When the device detects a frame on the channel, it reads the Duration value and updates its NAV timer. A device can only attempt to send data once its NAV timer reaches zero. However, in some cases, if multiple devices detect that the channel is free at the same time they may all try to transmit simultaneously, leading to collision.

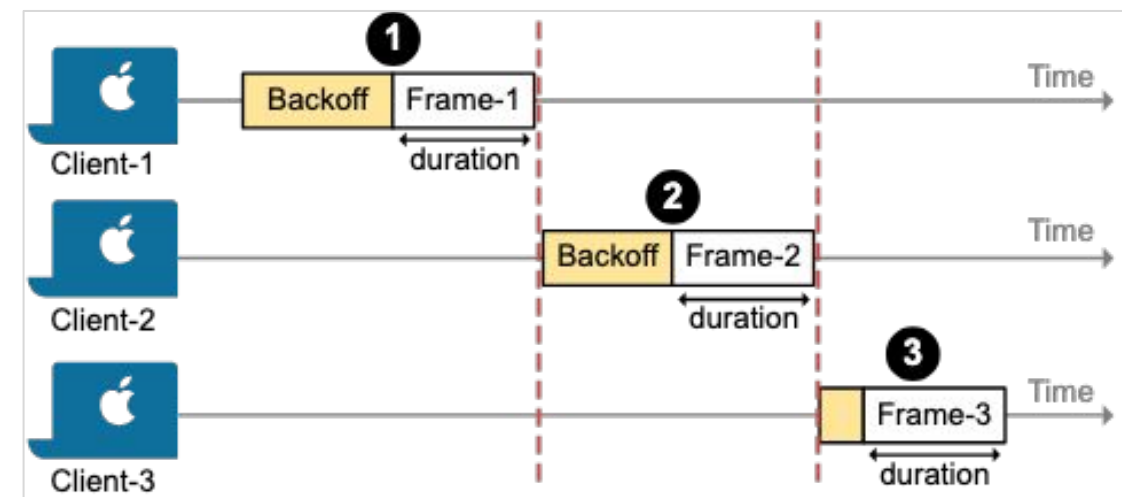


Collision Avoidance and Frame Acknowledgements

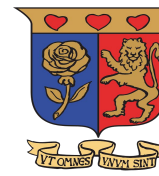


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- Wired devices detect collisions in real time and back off before retrying while wireless devices prevent collisions instead because they operate in half-duplex mode. They cannot detect a collision while transmitting.
- Mobile clients use a back-off timer when a client wants to transmit, by selecting a random number of timeslots between 0 and 31. This reduces chances of multiple clients transmitting at the same time. Only when the NAV and the backoff timers expires to zero, can the client compete for airtime. This is a simplified explanation of the backoff process it involves more timers and processes.
- Consider a client transmitted an 802.11 frame and assume that all collision avoidance methods worked correctly. Wireless uses collision avoidance, not collision detection. How does a client know its frame was received correctly?
- Since a device cannot listen while transmitting, it relies on a simple feedback system. When a client receives a unicast frame, it must send back a unicast acknowledgment (ACK) to confirm successful delivery.
- The 802.11 standard requires an ACK for each frame, except in 802.11n and 802.11e (WMM), where one acknowledgment can confirm the receipt of multiple frames in a burst

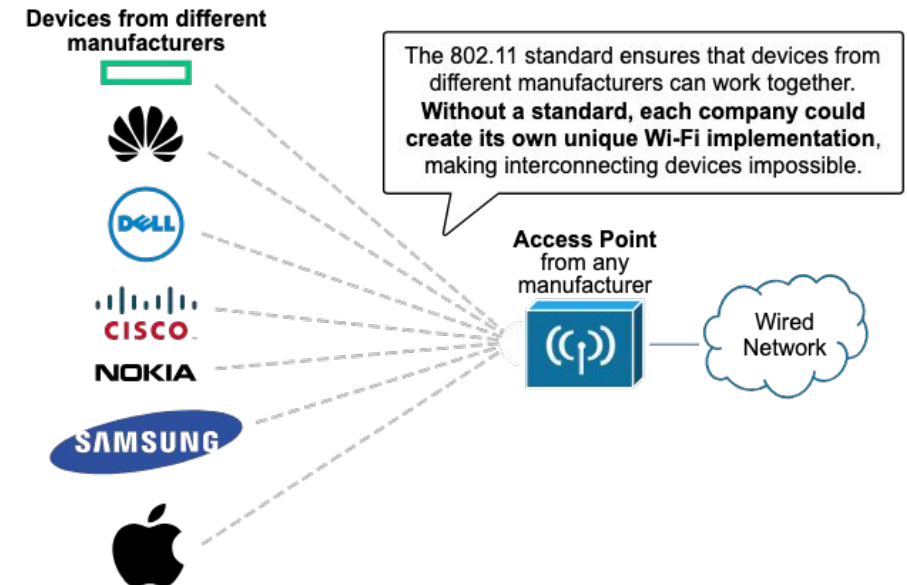


Standards and Regulatory Bodies

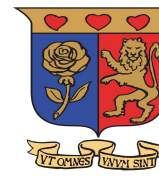


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- A standard is a set of rules, guidelines, or specifications designed to ensure that things follow the same rules and work consistently e.g. Human language. Human language is a set of rules and specifications to exchange information.
 - Wireless devices also use a common standard to exchange information, even if they are manufactured by different vendors in different parts of the world.
 - Standards ensure that devices from different manufacturers can communicate and work together. Without standards, each company could create its own unique system, making it difficult or even impossible to connect devices.
-
- Recall that RF spectrum is considered a valuable national resource, whose use is strictly controlled. It is limited, and thus holds value, through licensing.
 - It also supports national security by enabling military and emergency communications hence the need for a regulatory body that controls the allocation to for technological growth and public safety.
 - Since wireless communications use the RF spectrum as a medium, the RF regulatory bodies also control the wireless spectrum usage. They allocate bands for each Wi-Fi standard, set power and emission limits, and monitor



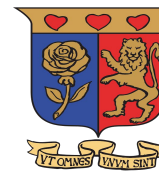
Standards and Regulatory Bodies



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- The regulatory bodies work parallel with IEEE to create technical standards, such as the IEEE 802.11 standards used in Wi-Fi.
- The IEEE as a global organization has many smaller “societies” that focus on different topics. For example, the IEEE Computer Society works on standards for networking, including Ethernet and Wireless LANs. They create standards through working groups. Each group has an open membership and is assigned a number. This number is added to the 802 family. For example, 802.11 is the group responsible for the wireless LAN standards used by all Wi-Fi vendors. Other working groups include:
 - 802.1: Network bridging (includes Spanning Tree Protocol)
 - 802.2: Link-layer control
 - 802.3: Ethernet
 - 802.11: Wireless LANs
 - 802.15: Wireless personal-area networks (like Bluetooth and ZigBee)

WIFI Standards



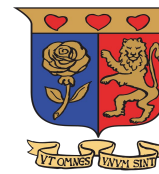
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- When technology changes, study groups (SGs) explore whether updates are needed. If so, task groups (TGs) work on amendments to the 802.11 standard. Each amendment is given a suffix, like 802.11a, 802.11b, and so on. If more are needed, two-letter suffixes are used, such as 802.11aa and 802.11aq.
- Sometimes, amendments take a long time to be approved. Manufacturers may release devices based on draft amendments such as with 802.11n, leading to some “Draft N-compliant” products that were not always compatible with others.
- The 802.11 standards often include the year they were approved, like 802.11-1997 or 802.11a-1999. Every few years, the IEEE revises the 802.11 standard to combine all previous amendments into one document. This keeps everything in one place.
- Even after amendments are combined, their names remain common references for the features they added.
- IEEE provides the technical framework for how wireless communication should work, while regulatory bodies manage the spectrum and enforce rules to ensure that these technologies can operate without causing interference. In short, IEEE is on the technical side of things, while the regulatory bodies are on the governing

IEEE Standard	WiFi Gen	Year	Frequency	Max Data Rate	Max Range
802.11		1997	2.4 GHz	2 Mbps	20m indoor 100m outdoor
802.11 a		1999	5 GHz	54 Mbps	35m indoor 120m outdoor
802.11 b		1999	2.4 GHz	11 Mbps	35m indoor 140m outdoor
802.11 g		2003	2.4 GHz	54 Mbps	38m indoor 140m outdoor
802.11 n	WiFi 4	2009	2.4/5 GHz	600 Mbps	70m indoor 250m outdoor
802.11 ac	WiFi 5	2013	5 GHz	6.9 Gbps	35m indoor
802.11 ad		2012	60 GHz	8.1 Gbps	3.3m indoor
802.11 ah		2017	Sub 1 GHz	347 Mbps	1km outdoor
802.11 ax	WiFi 6	2021	2.4/5/6 GHz	9.6 Gbps	30m indoor 120m outdoor
802.11 ay		2021	60 GHz	303 Gbps	10m indoor 100m outdoor
802.11 be	WiFi 7	2024	2.4/5/6 GHz	46.1 Gbps	30m indoor 120m outdoor

WIFI Standards

Wireless Security: Security Attacks

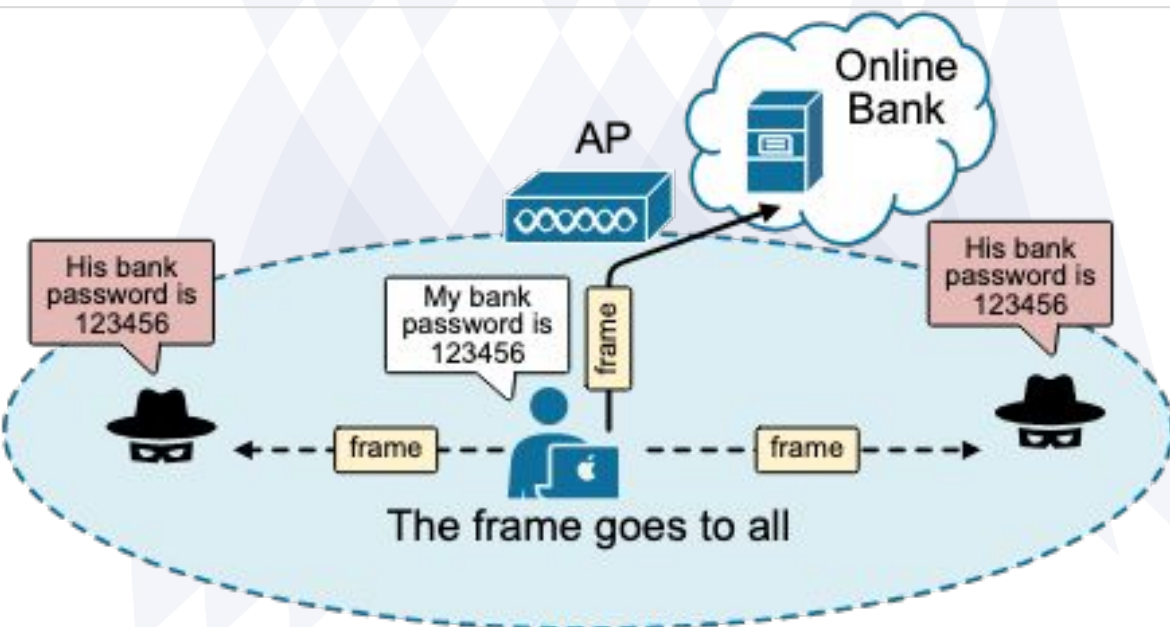


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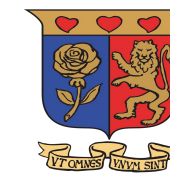
- Wired networks have an inherited level of security because data travels over physical wires, reducing the risk of eavesdropping. Wireless networks are different, anyone within range can potentially listen in. some of the most common security attacks that target Wi-Fi networks. Include:

Eavesdropping

- In Wi-Fi networks, clients share the same channel and coexist in the same coverage area. However, not all wireless devices are secure or trustworthy. Unlike wired networks, where data travels directly between sender and receiver, wireless signals spread in all directions, reaching any device within range.
- In this scenario, a user connects to his remote banking provider and shares his credentials. Due to the nature of Wi-Fi communication, the signal goes to the access point and to two untrusted users who capture the user's data.
- The ease of wireless communication also makes it vulnerable to eavesdropping and attacks. Therefore, strong encryption and integrity protection are paramount for this type of communication.

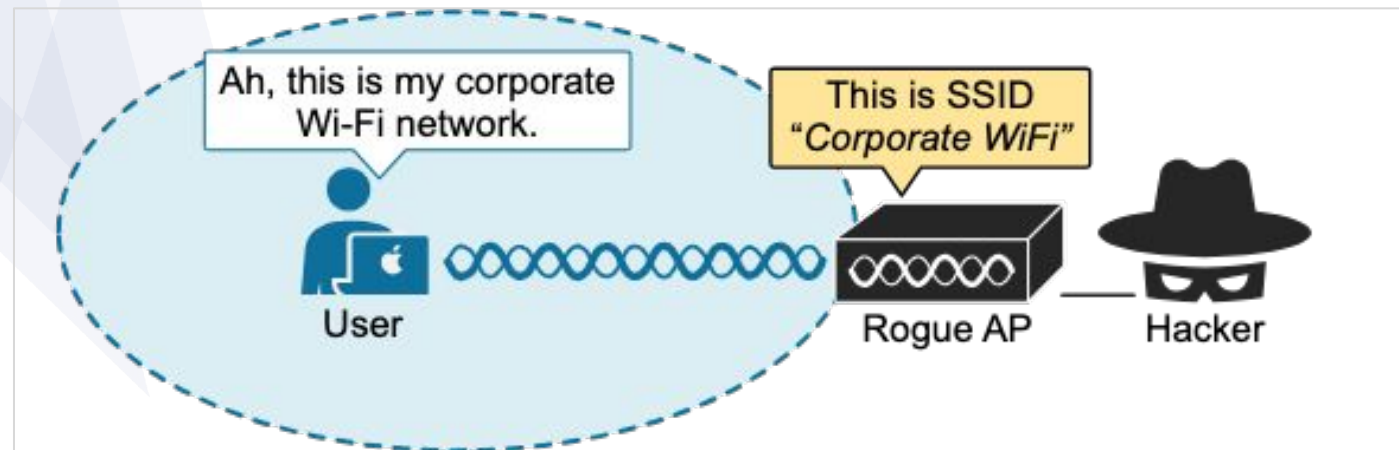


Wireless Security: Security Attacks-Evil Twin (Rogue AP)

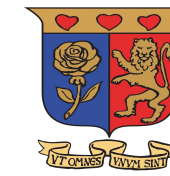


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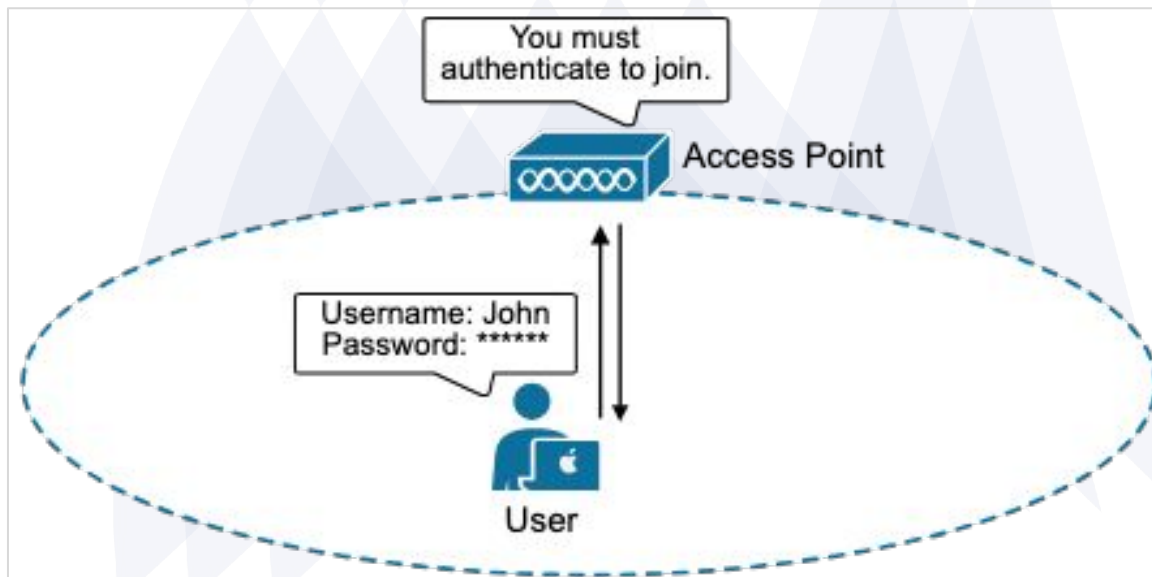
- When you join a Wi-Fi network, you usually focus on logging in, assuming the access point (AP) is trustworthy. For example, at the office, you see your corporate Wi-Fi name and connect without hesitation. But how do you know the AP is trustworthy?
- The only clue you have is the SSID name being broadcast. If it looks familiar, you might connect automatically. Your device may even join without asking if it's set to remember the network. But what if an attacker is faking that SSID?
- As shown, one of the most common wireless security concerns involves an attacker setting up a fake access point (AP) with the same SSID as a legitimate corporate network. The rogue AP sends beacons, responds to probes, and accepts client connections just like a legitimate AP.
- Users unknowingly connect, allowing the attacker to steal data. A fake AP can even send spoofed management frames to disconnect users from the real network and connect them to the fake SSID.



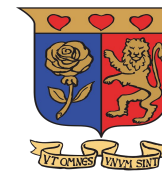
Wireless Security: Authentication



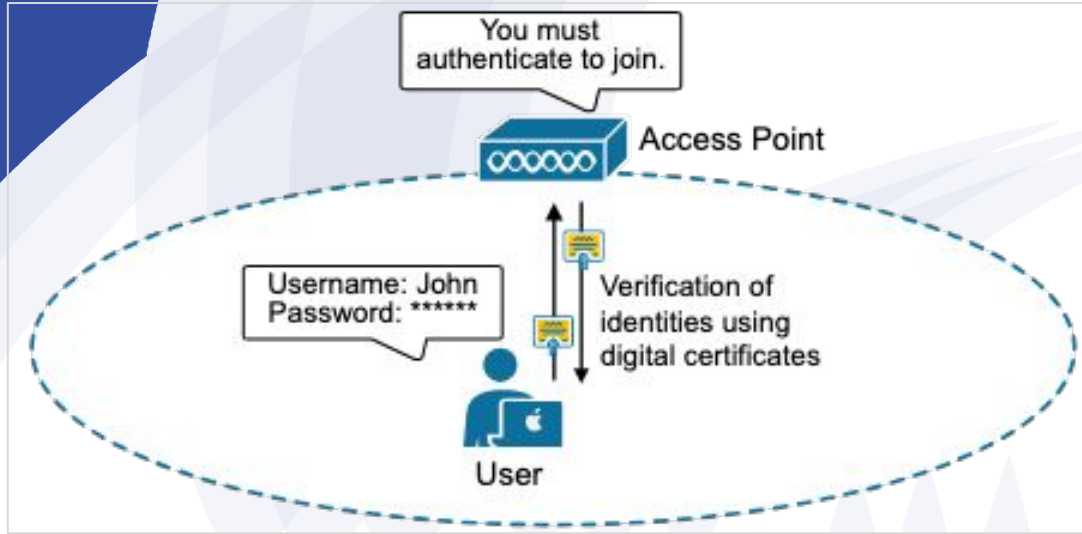
- For wireless clients to connect to a network, they must first find an SSID in range and request to join. Before they can connect, they need to be authenticated. *But why authentication is so important?*
- If your network connects to sensitive company resources, only trusted devices should be allowed in. Guest users, if permitted, should connect to a separate guest network with limited access. Rogue clients—unknown or unwanted devices—should be blocked completely.
- To control access, wireless networks authenticate devices before allowing them to connect. Clients must provide credentials to the access point (AP). The diagram shows a simple authentication process between a user and an AP:



Wireless Security: Authentication



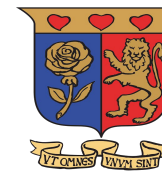
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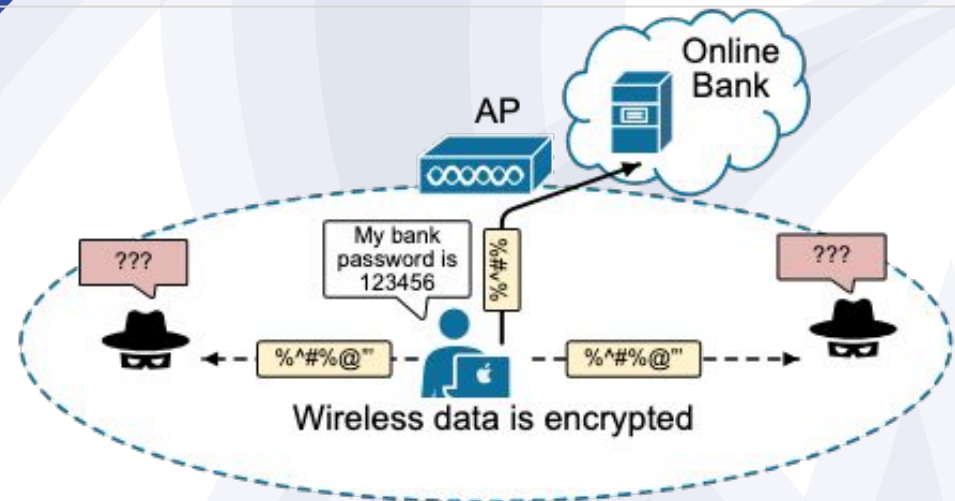
- Different authentication methods exist; Some use a shared password stored on all trusted devices. But if a device is lost or stolen, anyone could use it to connect.
- More secure methods check credentials against a company database, requiring a valid username and password that threat actors wouldn't know.
- APs broadcast SSIDs, and if the name looks familiar, you might connect automatically. Some devices even auto-connect to known SSIDs without asking. Attackers can set up fake APs that mimic real ones, tricking devices into connecting. Once a client joins, the attacker can intercept data or disrupt connections by sending spoofed management frames.
- To prevent such attacks, clients should verify APs before connecting. The diagram show how this works:

Clients should also authenticate management frames to ensure they come from a trusted AP.

Wireless Security: Message Privacy

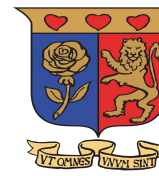


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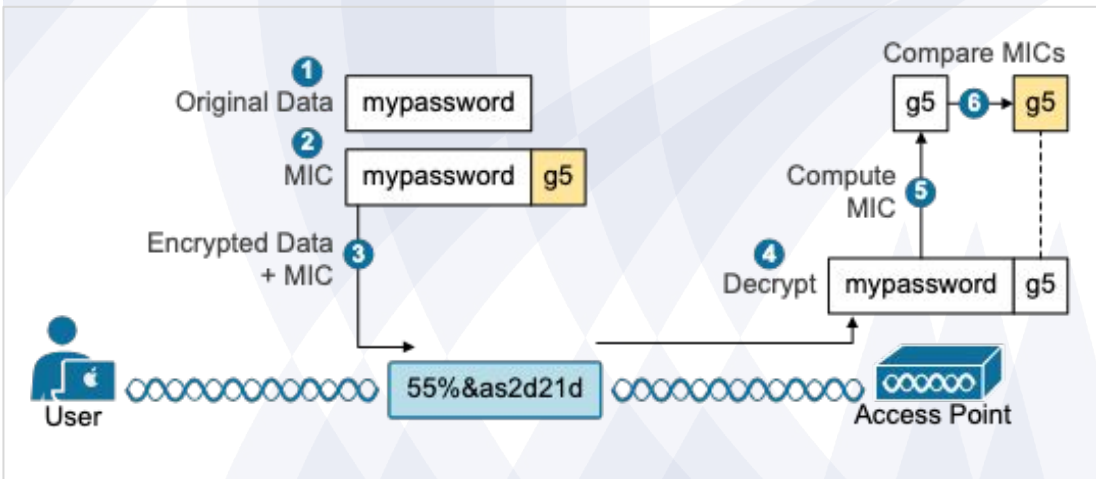
- If a client must authenticate before joining a Wi-Fi network, it can also verify the AP and its management frames. This builds trust, but data is still exposed to eavesdroppers on the same channel.
- To keep data private, wireless networks use encryption. The data in each frame is encrypted before transmission and decrypted upon arrival. Both the sender and receiver must use the same encryption method.
- Each WLAN supports only one authentication and encryption scheme, so all clients must use the same encryption method. However, this doesn't mean every client can see others' data since the AP assigns a unique encryption key to each client.
- Ideally, only the AP and the client share the encryption key, preventing others from eavesdropping. In the diagram, a client's password is encrypted before transmission. The AP decrypts it before sending it to the wired network, while other devices cannot read it.
- The AP also has a group key for sending encrypted data to all connected clients. Each client uses this key to decrypt broadcast messages from the AP

Wireless Security: Message Integrity

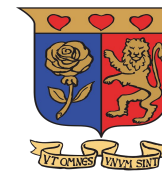


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- Encryption hides data while it travels over an untrusted network. The recipient can decrypt it, but what if the data is altered along the way? Detecting changes would be difficult.
- A Message Integrity Check (MIC) helps prevent tampering. Think of it as a secret stamp added inside the encrypted data frame.
- The stamp is based on the data itself. When the recipient decrypts the frame, it calculates its own stamp and compares it to the original. The diagram shows a simplified overview of the process.
- If both stamps match at step six, it is mathematically proven that the data is intact and has not been altered along the way. If not, the data was changed in transit.

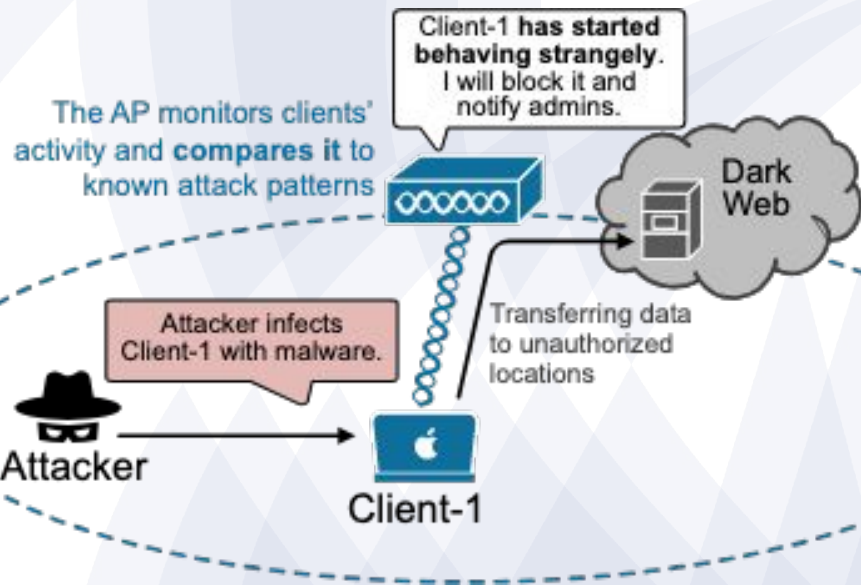


Wireless Security: Intrusion Protection

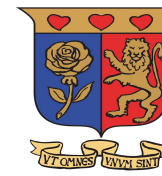


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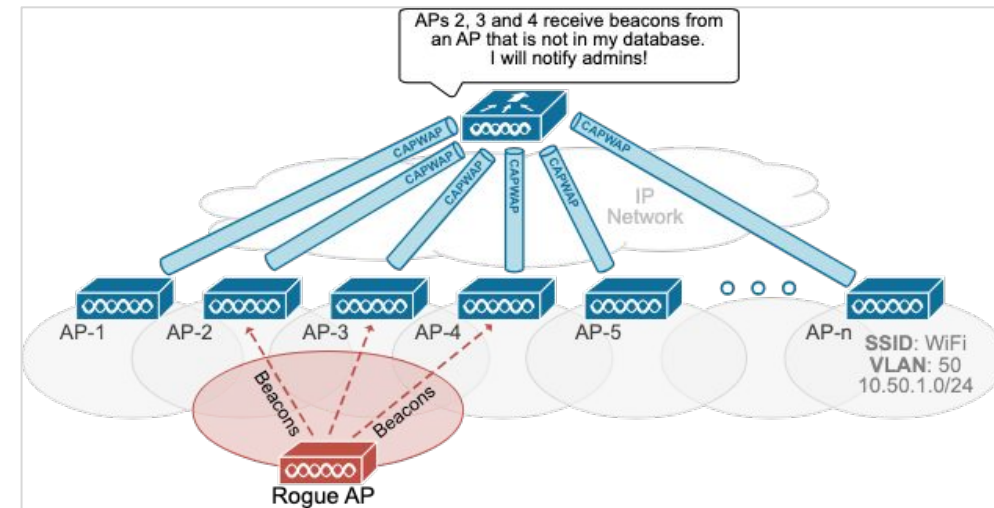
- Most wireless security measures focus on blocking attackers from joining the network, serving as perimeter security. However, what if a client who has already joined the network gets malware and starts engaging in malicious activities?
- Access points should monitor clients' activity and compare it to known attack patterns. APs can detect some threats and alert their controllers and network admins, as shown.
- From a broader perspective, the wireless controller has a better and more accurate view of the Wi-Fi domain. It can analyze data from multiple APs, better detect malicious activities, and apply IPS policies to clients.



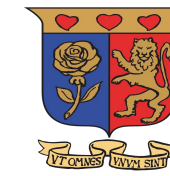
Wireless Security: Controller-assisted Security



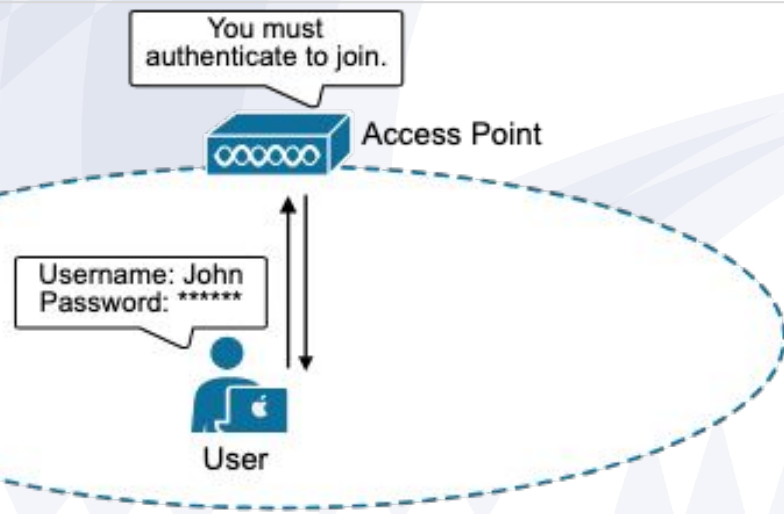
- Regardless of how well secured the network is, it is possible to introduce a rogue own access point and connect it to your wired network or just start broadcasting your corporate SSID. This can cause interference or perform malicious activities.
- Any devices that connect illegally to your network or broadcast your SSID are rogue APs and clients that connect to them are rogue clients.
- Controllers should identify both since the WLC controller has a high-level view of the network. It can listen for beacon signals from APs. It uses detection algorithms to check if an AP is in the network's database. If not, it is classified as rogue, as shown in the diagram below. As an administrator, you can manually mark an AP as friendly or rogue.
- The controller can also send special test frames to check if a rogue AP is connected. If the AP receives these frames and forwards them, it confirms that a rogue AP is connected, and you can remediate this.
- The controller can also block rogue APs, it monitors wireless clients that connect to a rogue AP and sends fake de-authentication signals to disconnect them, making them believe the rogue AP has removed them.
- The wireless controller can detect many types of attacks; passive attackers secretly capturing wireless data or active trying to disrupt the network.
- One example is an attacker flooding an AP with fake connection requests, causing it to crash. Another attack is like rogue containment, where an attacker sends fake de-authentication signals to kick real users off the network repeatedly.



Wireless Security: Client Authentication Methods



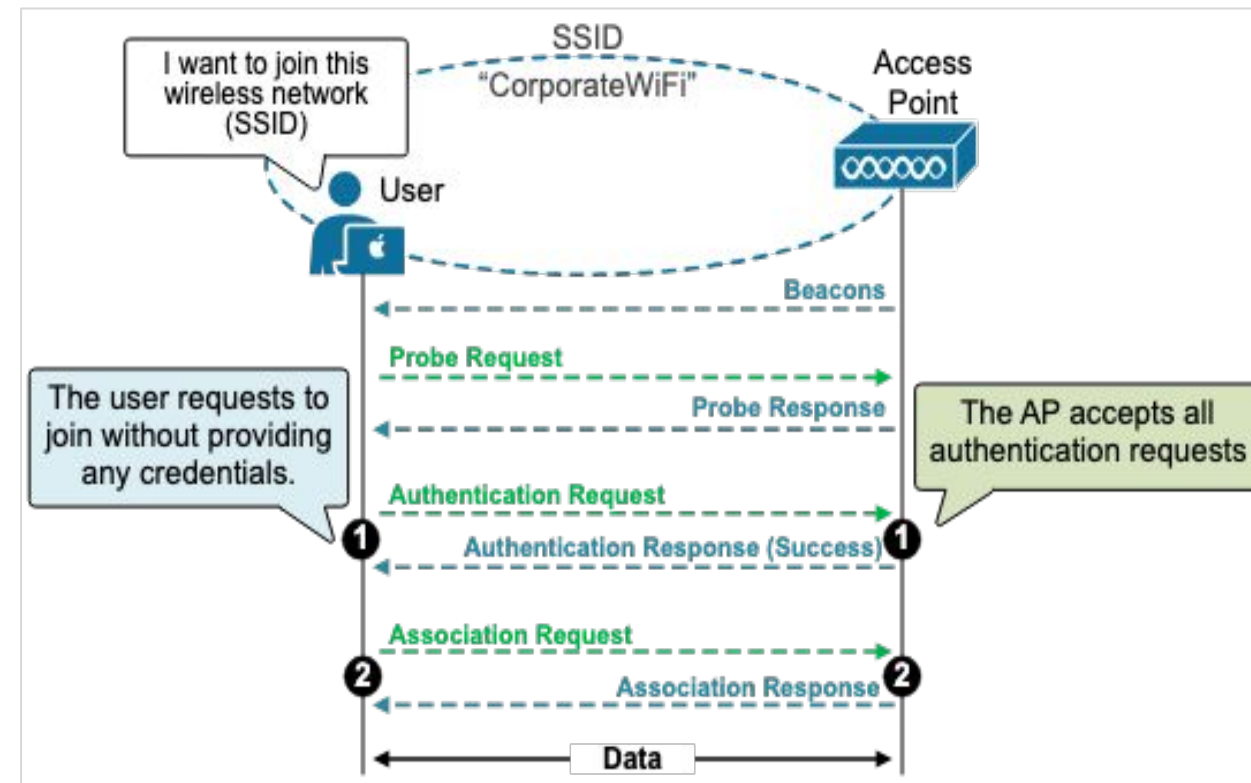
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- The original 802.11 Wi-Fi standard defined two basic authentication methods:
 - Open System Authentication (OSA)
 - Shared Key Authentication (SKA)
- Wireless security introduces a lot of short acronyms and terms. The most common wireless client authentication methods and technologies includes terms like WEP, OSA, SKA, PSK, TKIP, EAP, WPA, WPA2, and WPA3. Understanding the concept behind an acronym is vital and helps remember the concept.
- Authentication is the process of verifying the identity of a device before allowing it to connect to the network. It ensures that only authorized users and devices can access the network and its resources.
- Authentication is important because wireless networks are more vulnerable to unauthorized access if there is no verification system in place.

Open System Authentication (OSA)

- Open authentication allows unrestricted access to a wireless network; no authentication is required-any device can connect. This is usually found in public Wi-Fi networks.
- The only step required is for the client to send an 802.11 authentication request before associating with an access point (step 1 in the diagram below). No password or credentials are needed.
- Open authentication is not secure because any client can connect without being challenged.
- Sometimes, open web authentication is added for basic screening. In this case, clients can connect to the WLAN but must open a browser, accept the terms, and enter email before gaining full access.
- Most operating systems warn users that open networks are not secure and that you should not access sensitive information.

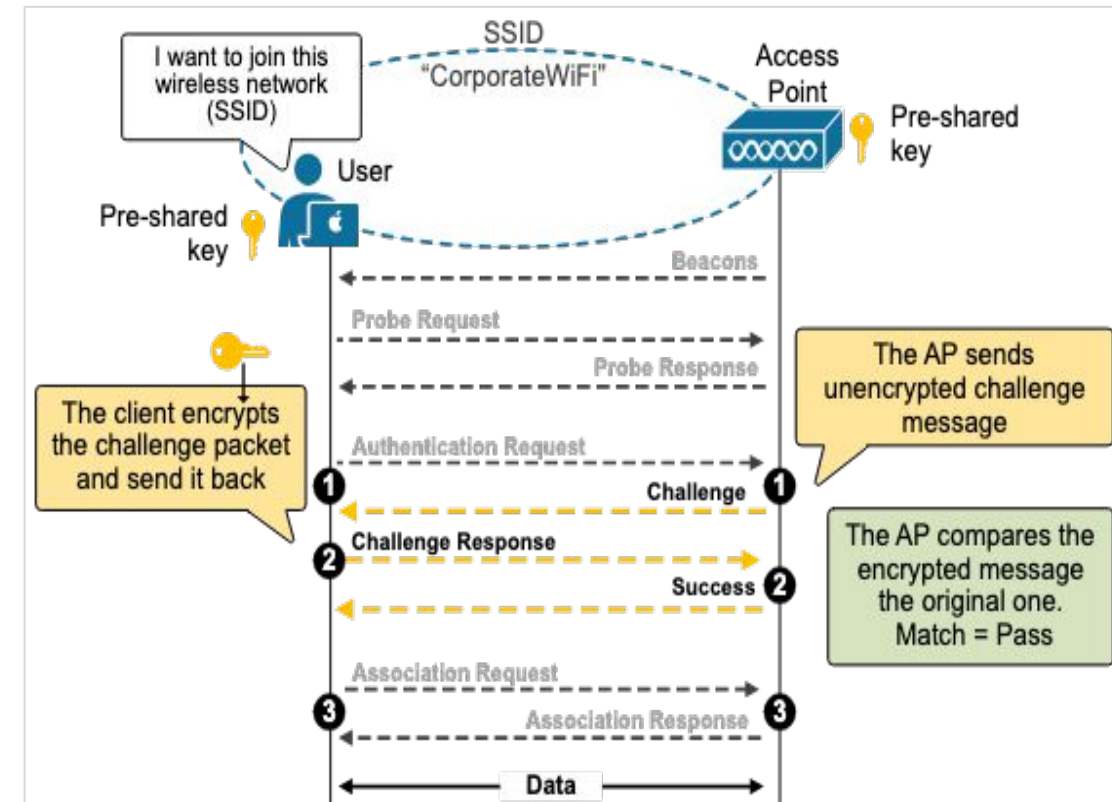


Client Authentication Methods: Shared Key Authentication (SKA)- WEP

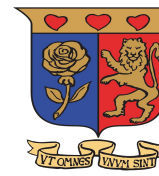


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- The Open System Authentication (OSA) is basically a technique to force users to access the terms of service of a public SSID. Wireless Equivalent Privacy (WEP) is introduced to improve security. It aimed to make wireless connections as secure as wired ones (hence the name).
 - WEP uses a static, shared key that must be manually configured on both the AP and the wireless client. The key (the password) must be shared with the user ahead of time (hence the name shared key). See the diagram (Notice that both AP and device have the pre-shared key distributed by admin)
-
- Step 0: After discovering the AP (via beacons and probes), the client sends an authentication request to join the network.
 - Step 1: The access point responds with a challenge. The challenge message is unencrypted.
 - Step 2: The client encrypts the message using the pre-shared key and transmits it wirelessly back to the AP.
 - Success: The AP receives the challenge response, uses the same shared key, decrypts the message, and compares it to the original one. If both match, the AP grants access to the client.
 - Once the user is authenticated and joins the network, WEP uses the RC4 cipher to encrypt and decrypt data, preventing eavesdropping. Both sender and receiver use the same key, called a **WEP key**, to generate unique encryption keys for each data frame. As long as both devices have the same WEP key, they can encrypt and decrypt each other's data.



Client Authentication Methods: Why WEP is Insecure

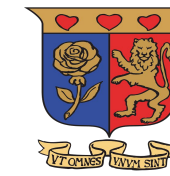


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- WEP keys can be 104 bits long. In theory, longer keys provide stronger encryption. However, WEP has known weaknesses. By 2001, security flaws were discovered, and by 2004, WEP was deprecated when the improved 802.11i amendment was adopted.
- Even though WEP is outdated and insecure, it remained in use for many years. This is because new security methods required new hardware with cost implications. As a result, WEP continued to be supported for backward compatibility and can still be found in some devices today. Common WEP vulnerabilities and exploits include:
 - Weak encryption algorithm (RC4).
 - Predictable Initialization Vectors (IVs), making it easy for attackers to crack the key.
 - Replay attacks and eavesdropping risks.
- Because of these issues, WEP has been deprecated, and more secure authentication methods like WPA2 and WPA3 are recommended instead.

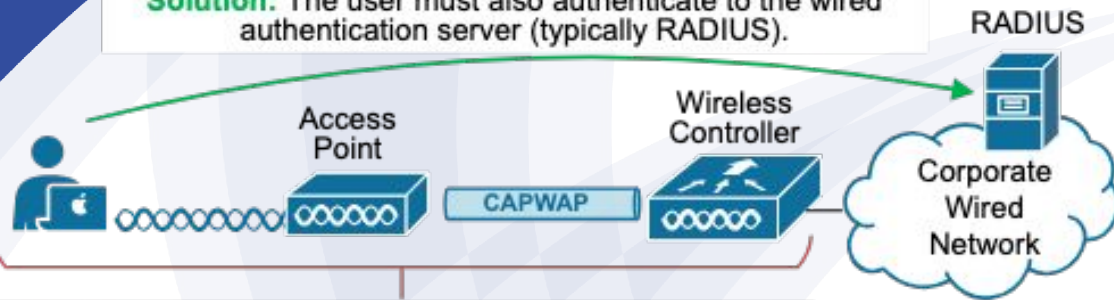
Client Authentication Methods.

802.1x/Extensible Authentication Protocol (EAP)

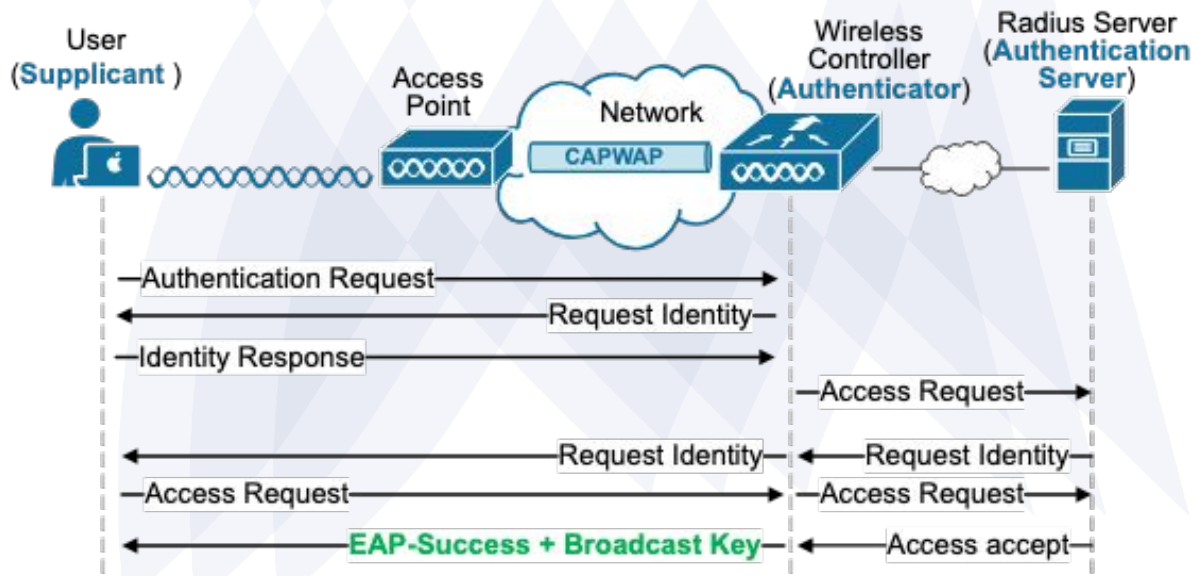


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Solution: The user must also authenticate to the wired authentication server (typically RADIUS).

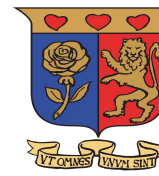


Problem: Once the user successfully authenticates and joins the SSID, it directly gets access to the wired network.

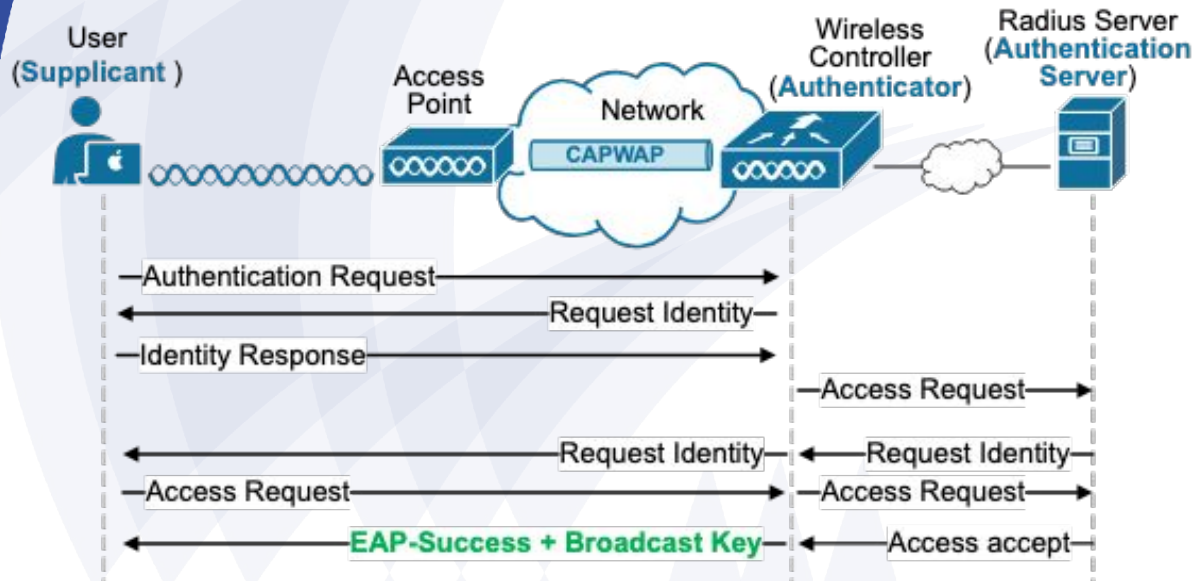


- The 802.11 open system authentication and WEP (shared key authentication), were not very secure. A better way to handle authentication was needed.
- Client authentication usually involves a challenge, a response, and then a decision to allow or deny access. In the background, the process may also exchange encryption keys and other settings needed for the client to connect.
- The standard authentication methods have one major inefficiency - once a user authenticates successfully with the AP, it can directly access the wired network, as shown(graphic 1).
- To address this, **Extensible Authentication Protocol (EAP)** was introduced, adding an additional layer of security to the authentication process and controls network access based on IEEE 802.1x .
- When 802.1x is used, even if a client successfully authenticates to the wireless network, it still cannot access the wired network until it successfully authenticates with the wired authentication server (typically RADIUS) (graphic 2).

Client Authentication Methods: 802.1x/EAP



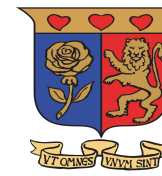
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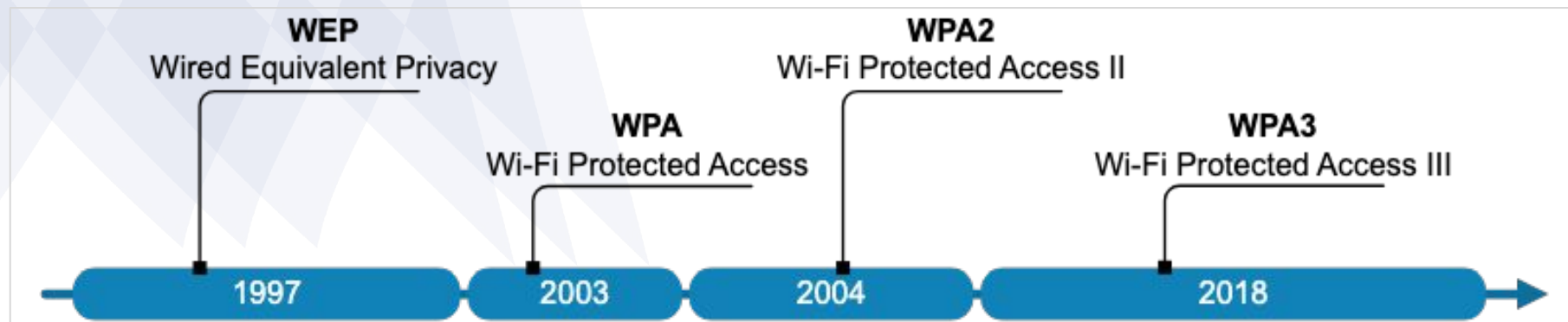
- The wireless controller and the Radius server work together to authenticate the user against the organization's user database.
- The WLC acts as a bridge between the client and the authentication server, using EAP to manage user authentication and access control.

- The diagram shows a very simplified example of 802.1x-EAP authentication. Notice that, there are three key roles as follows:
 - **Supplicant** – The client device that asks for network access.
 - **Authenticator** – A network device, like a wireless LAN controller (WLC), that manages access.
 - **Authentication Server (AS)** – A server (usually a RADIUS server) that checks the client's login details and allows or blocks access based on stored user data and policies.

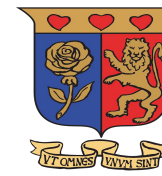
Client Authentication Methods: Wi-Fi Protected Access (WPA, WPA2, WPA3)



- EAP introduced a great additional layer of security by incorporating 802.1x. Still, it did not include robust encryption by itself and had to rely on WEP (Wired Equivalent Privacy), which later proved to be insecure due to known vulnerabilities.
- To address these security weaknesses, the Wi-Fi Alliance introduced Wi-Fi Protected Access (WPA). The first version of WPA used a key management protocol called TKIP (Temporal Key Integrity Protocol). It relied on the weaker RC4 encryption algorithm and was quickly proven to be unsecure.
- The Wi-Fi Alliance introduced WPA2 as the next generation of wireless security. WPA2 replaced TKIP with the stronger AES (Advanced Encryption Standard) using the CCMP algorithm for encryption.
- Like WPA, WPA2 supported 802.1x-EAP for enterprise-grade security while also offering a simpler "Personal Mode" using a pre-shared key (PSK) for smaller SOHO deployments.
- The diagram shows the timeline of the major Wireless security protocols.

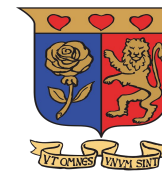


Client Authentication Methods: Wi-Fi Protected Access (WPA, WPA2, WPA3)



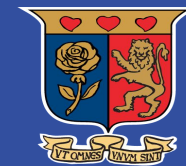
- It is important to understand that every Wi-Fi Protected Access (WPA) protocol offers two types of authentication:
 - Personal Mode: Uses a pre-shared key (PSK) to authenticate clients.
 - Enterprise Mode: Uses 802.1x and EAP-based authentication for better security.
- The personal mode uses a shared password, which can be a single passphrase easily shared with users. It is ideal for small deployments.
- The enterprise mode relies on an organizational RADIUS server for authentication. It is designed for large-scale organizations that have a centralized database to authenticate the clients against.
- WPA security protocols use encryption and session key algorithms with varying complexity levels as shown in the table (next slide). It compares each of the major WiFi security protocols and introduces the primary facts that you should remember.

Wi-Fi Protected Access (WPA, WPA2, WPA3)



	WEP	WPA	WPA2	WPA3
Released	1997	2003	2004	2018
Encryption protocol	RC4	TKIP	CCMP (with AES)	GCMP (with AES)
Session Key	40 or 104bits	128bits	128bits	128/256bits
Authentication	Open system (OSA)/ Shared Key (SKA)	802.1X / Pre-shared Key	802.1X / Pre-shared Key	SAE (Simultaneous Authentication of Equals)
Level of security	Low	Medium	High	Very High
Vulnerabilities	Easily cracked due to weak RC4	Vulnerable to TKIP exploits	PSK vulnerable to brute force	Improved against DDoS, Brute Force and MITM attacks

- From the table that WPA3 is currently the most secure wireless protocol. However, note that wireless security isn't a one-time task that is single-dimensional. Every organization must constantly follow the best practices like regular updates, enforcing strong passwords, and continuous monitoring.



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